

Supplementary Information

Generated by `src_v8/make_table_s1.py` from `results_pub/` (Sections S3 and S8 from `results_v8/`). $\tau = 5\%$.

S1. Per-run reliability accounting

Taxonomy columns (T1–T4, modes) count all 20 committed geometries of a run, including the solver-unresolved design of A at seed 777; `n_valid` and the pretender, oracle-confirmed and Δ -flagged counts exclude it.

Table S1: S1. Per-run reliability accounting

S	seed	n_valid	surrogate-claimed [Wilson 95 %]	oracle-confirmed [Wilson]	pretenders [Wilson]	Δ -flagged [Wilson]	flag thr (%)	modes	T1	T2	T3	T4	tail RSD	endpoint spread (u)	ρ (RCWA, Surr) [CI95]	p
A	42	20	20 [83.9, 100.0]	19 [76.4, 99.1]	1 [0.9, 23.6]	2 [2.8, 30.1]	3.51	20	16	14	0	0	0.0102	1.832	+0.34 [-0.13, 0.69]	0.143
A	123	20	20 [83.9, 100.0]	20 [83.9, 100.0]	0 [0.0, 16.1]	3 [5.2, 36.0]	3.55	19	15	18	2	0	0.0124	1.819	+0.45 [-0.01, 0.76]	0.044
A	777	19	19 [83.2, 100.0]	18 [75.4, 99.1]	1 [0.9, 24.6]	2 [2.9, 31.4]	3.35	20	13	14	0	0	0.004	1.882	+0.19 [-0.30, 0.59]	0.446
B	42	20	20 [83.9, 100.0]	17 [64.0, 94.8]	3 [5.2, 36.0]	7 [18.1, 56.7]	3.44	16	17	15	7	0	0.0059	1.595	+0.12 [-0.34, 0.53]	0.622
B	123	20	20 [83.9, 100.0]	17 [64.0, 94.8]	3 [5.2, 36.0]	4 [8.1, 41.6]	3.15	12	12	14	11	0	0.0053	1.547	+0.15 [-0.32, 0.55]	0.539
B	777	20	20 [83.9, 100.0]	17 [64.0, 94.8]	3 [5.2, 36.0]	5 [11.2, 46.9]	3.36	16	12	12	7	0	0.011	1.537	-0.12 [-0.53, 0.34]	0.618
C	42	20	20 [83.9, 100.0]	18 [69.9, 97.2]	2 [2.8, 30.1]	1 [0.9, 23.6]	4.21	17	1	9	5	0	0.0015	1.409	+0.75 [0.41, 0.91]	1.4e-04
C	123	20	20 [83.9, 100.0]	19 [76.4, 99.1]	1 [0.9, 23.6]	0 [0.0, 16.1]	4.21	15	1	11	9	0	0.0045	1.356	+0.83 [0.56, 0.94]	6.4e-06
C	777	20	20 [83.9, 100.0]	18 [69.9, 97.2]	2 [2.8, 30.1]	0 [0.0, 16.1]	4.08	17	0	9	6	0	0.002	1.431	+0.72 [0.36, 0.89]	3.6e-04
all		179			16/179 (A 2/59, B 9/60, C 5/60)											

S2. Commitment-rule comparison (seed 42)

Table S2: S2. Commitment-rule comparison (seed 42)

S	rule	surrogate-claimed	oracle-confirmed	pretenders (of n)	pretenders / claimed [Wilson]	mean oracle MAE (%)	Fisher p vs best-of-8 (design-level)	Fisher p (conditional on claimed)
A	gradient best-of-8	20	19	1/20	1/20 [0.9, 23.6]	2.57	—	—
A	random search (budget-matched)	20	16	4/20	4/20 [8.1, 41.6]	4.73	0.342	0.342
A	single start (mid-box)	19	15	4/20	4/19 [8.5, 43.3]	3.30	0.342	0.182
B	gradient best-of-8	20	17	3/20	3/20 [5.2, 36.0]	3.74	—	—
B	random search (budget-matched)	20	14	6/20	6/20 [14.5, 51.9]	5.43	0.451	0.451
B	single start (mid-box)	19	15	4/20	4/19 [8.5, 43.3]	3.81	1.000	0.695
C	gradient best-of-8	20	18	2/20	2/20 [2.8, 30.1]	1.85	—	—
C	random search (budget-matched)	16	15	1/20	1/16 [1.1, 28.3]	4.07	1.000	1.000
C	single start (mid-box)	15	11	4/20	4/15 [10.9, 51.9]	3.96	0.661	0.367

S3. Order-5 vs order-7 reference-solver MAE per committed design (seed 42; as-submitted arm, results_v8/)

Table S3: S3. Order-5 vs order-7 reference-solver MAE per committed design (seed 42; **as-submitted arm**, results_v8/)

S	#	Surr MAE (%)	RCWA MAE order 5 (%)	RCWA MAE order 7 (%)	shift (pp)	pass@5	pass@7	status
A	1	0.07	1.74	0.36	-1.38	1	1	stable
A	2	0.56	2.84	3.95	+1.12	1	1	stable
A	3	1.22	2.74	2.94	+0.20	1	1	stable
A	4	0.05	2.54	2.64	+0.10	1	1	stable
A	5	0.17	7.56	7.58	+0.02	0	0	stable
A	6	0.47	7.42	7.00	-0.42	0	0	stable
A	7	0.47	3.26	3.45	+0.19	1	1	stable
A	8	0.27	2.90	3.59	+0.69	1	1	stable
A	9	0.21	5.83	5.99	+0.16	0	0	stable
A	10	0.63	6.76	6.26	-0.51	0	0	stable
A	11	1.24	4.44	4.78	+0.35	1	1	stable
A	12	0.45	3.06	3.80	+0.74	1	1	stable
A	13	1.00	4.26	5.25	+0.99	1	0	pass→fail
A	14	0.21	6.50	7.52	+1.03	0	0	stable
A	15	1.43	6.48	7.00	+0.52	0	0	stable
A	16	0.37	6.73	6.24	-0.49	0	0	stable
A	17	0.60	4.22	4.27	+0.04	1	1	stable
A	18	1.46	7.36	8.31	+0.95	0	0	stable
A	19	0.20	5.92	5.78	-0.14	0	0	stable
A	20	0.08	4.36	4.35	-0.00	1	1	stable
B	1	0.45	1.84	2.39	+0.55	1	1	stable
B	2	1.14	2.60	3.67	+1.07	1	1	stable
B	3	0.76	3.80	2.22	-1.58	1	1	stable
B	4	0.43	3.76	7.76	+3.99	1	0	pass→fail
B	5	1.07	1.96	4.40	+2.44	1	1	stable
B	6	1.10	1.62	3.42	+1.80	1	1	stable
B	7	0.61	31.64	26.10	-5.54	0	0	stable
B	8	0.63	5.86	3.31	-2.55	0	1	fail→pass
B	9	0.70	2.78	1.92	-0.86	1	1	stable
B	10	1.88	6.82	5.38	-1.44	0	0	stable

S	#	Surr MAE (%)	RCWA MAE order 5 (%)	RCWA MAE order 7 (%)	shift (pp)	pass@5	pass@7	status
B	11	1.03	2.51	7.20	+4.69	1	0	pass→fail
B	12	0.35	7.58	2.97	-4.62	0	1	fail→pass
B	13	0.30	2.56	3.46	+0.90	1	1	stable
B	14	0.30	7.54	6.05	-1.49	0	0	stable
B	15	0.31	1.49	1.97	+0.48	1	1	stable
B	16	1.22	7.25	3.87	-3.38	0	1	fail→pass
B	17	0.37	10.74	9.73	-1.00	0	0	stable
B	18	0.67	13.40	9.30	-4.10	0	0	stable
B	19	0.49	3.53	7.07	+3.54	1	0	pass→fail
B	20	1.07	13.68	14.96	+1.28	0	0	stable
C	1	2.57	9.15	6.96	-2.19	0	0	stable
C	2	0.50	3.60	17.67	+14.07	1	0	pass→fail
C	3	2.16	3.79	4.61	+0.83	1	1	stable
C	4	1.75	3.42	4.05	+0.64	1	1	stable
C	5	1.09	3.37	8.43	+5.06	1	0	pass→fail
C	6	1.75	3.80	3.88	+0.08	1	1	stable
C	7	3.88	5.59	5.84	+0.25	0	0	stable
C	8	0.42	5.72	7.80	+2.08	0	0	stable
C	9	1.36	6.87	9.65	+2.78	0	0	stable
C	10	3.29	6.49	6.13	-0.37	0	0	stable
C	11	3.88	11.06	11.07	+0.02	0	0	stable
C	12	0.83	5.67	6.68	+1.01	0	0	stable
C	13	0.40	10.79	10.05	-0.74	0	0	stable
C	14	2.79	3.82	4.84	+1.02	1	1	stable
C	15	0.87	14.54	9.76	-4.78	0	0	stable
C	16	2.01	3.99	7.95	+3.96	1	0	pass→fail
C	17	3.62	5.70	6.29	+0.59	0	0	stable
C	18	2.76	9.78	10.57	+0.79	0	0	stable
C	19	2.13	7.92	13.92	+6.01	0	0	stable
C	20	0.43	11.93	6.51	-5.42	0	0	stable
totals						A 11→10 (flips 1) / B 11→11 (flips 6) / C 7→4 (flips 3)		

S4. Pre-specified protocol (verbatim)

Reproduced verbatim from the frozen record, including its internal headings (e.g. its Section 7), its then-current description of [1] as under review, and its phrase "independent codex review", which denotes AI-assisted internal review. Frozen 2026-06-10 (stated in the file) for the audit of the as-submitted release, and re-run unchanged on the published pipeline in September 2026 (manuscript Section 2.7); file last modified 2026-06-11 09:39 UTC. Two items post-date the freeze (manuscript Section 2.7): the restart-seeding sentence of §3.4, and the multi-seed parenthesis of §5 item 5, which was appended after the seed-42 runs; the seed-123/777 artifacts pre-date the file's last modification, so the record does not establish that the extension was written before those runs.

v8 Protocol — Reliability of Physics-Transferred Surrogates in Inverse Metasurface Design

Status. Pre-specified protocol, frozen 2026-06-10, before any v8 computation. Supersedes all v4a/v7 numerical results (invalidated by the wavelength-grid bug, the frozen-optimizer bug, and the Δ -coupling statistical artifact; see docs/review_iter4_full_audit.md). The protocol *specification* (metric hierarchy, taxonomy, thresholds) carries over from manuscript v4 §2.3–2.5 with the corrections listed in §6 below.

1. Research question

Paper 1 (Choi et al. 2026, under review) introduced a pilot-set-computable transferability diagnostic — the TMM–RCWA Pearson correlation r — and showed it predicts *forward* prediction benefit of physics-based transfer learning (PBTl). The present study asks:

Does r predict the trustworthiness of a physics-transferred surrogate when it is used as the forward block of a gradient-based inverse-design loop?

Hypothesis H1 (from Paper 1 §4.2 #4, §4.6 #3): downstream reliability degrades as r falls. Pre-specified orderings to test, across the three anchors A (median $r = +0.72$), C (+0.34), B (−0.07):

- H1a: RCWA-oracle success rate is non-increasing as r falls ($A \geq C \geq B$).
- H1b: Δ -flag rate is non-decreasing as r falls ($A \leq C \leq B$).
- H1c: $\rho(\text{RCWA MAE}, \text{Surr MAE})$ — "is the surrogate's self-report informative about true quality?" — is non-increasing as r falls.

Any outcome is reportable: confirmation, partial confirmation, or refutation (the last would itself be a finding: forward-transferability diagnostics do not transfer to the inverse setting).

2. Design

Three structures from the Paper 1 release, each run through the **identical** pipeline (single parameterized codebase, src_v8/):

Table S4: S4. Pre-specified protocol (verbatim) — protocol §2. Design

	Structure A	Structure B	Structure C
Physics	asym. dual-dielectric dual-cavity	ring-disk Fano	anisotropic rect (TE/TM)
Median pilot r (Paper 1)	+0.72	−0.07	+0.34
Design params d	10	8	7
Physics features	17	13	13
Wavelength grid (native)	380–780 nm $\times$ 100	400–1800 nm $\times$ 100	400–1800 nm $\times$ 100
Dataset	struct_A_vis_500	struct_B_500	struct_C_500
Good samples (filter §3.1)	479/500	461/500	400/500

	Structure A	Structure B	Structure C
Output channels	A,R	A,R	A_TE,R_TE,A_TM,R_TM
Surrogate	MPhys (1 head)	MPhys (1 head)	MPhys_dual (2 heads)

Wavelength grids are **per-structure native grids** (the grids the datasets and the TMM-pretrained checkpoints were built on). Cross-structure comparisons are of dimensionless quantities (rates, correlations, ratios), not raw spectra. This is disclosed; the 380–780 nm claim of manuscript v4 §2.2 was wrong for B/C and is retired.

3. Pipeline (per structure)

3.1 Data hygiene (mirrors Paper 1 exactly)

good = all-wavelength $A \geq -0.01$ (C: A_TE and A_TM), then clip A,R to [0,1]. Divergent torcwa rows (39 B / 100 C / 21 A) never enter training, validation, testing, or the target pool.

3.2 Split (mirrors Paper 1 stage 2 exactly)

`rng(42).permutation(N_good) → test = last 50, val = previous 50, remaining = rest`. Fine-tune train set = first 350 of `rng(42)`-permuted remaining (A: 350/379, B: 350/361, **C: full 300/300** — the upstream code trains on the full pool when it is smaller than 350 and we disclose `n_train(C) = 300`).

3.3 Surrogate fine-tune (faithful to the pretrained checkpoints)

Start from Paper 1's TMM-pretrained checkpoints (`pretrained_mphys_tmm{,_B,_C}.pt`), loaded **strict**. Faithfulness requirements (violations of these are what invalidated v7's "pretrained" claim):

1. Geometry input is [`wl_norm`, `params_norm`] (wavelength **first**).
2. `params_norm` uses the **training bounds** of the checkpoint: `A = get_bounds("A")`, `B = BOUNDS_B`, `C = BOUNDS_C with Wx,Wy ∈ [50,400]` (the C dataset is sampled wider, up to ~700 nm; the upstream fine-tune fed normalized values > 1 for those samples and we mirror that, disclosed).
3. Physics features are normalized with the **TMM-set statistics**, recomputed deterministically (features depend only on `rng(99)` TMM params and the wavelength grid — no simulation needed).
4. Loss = MSE(A) + MSE(R) (C: all four channels), AdamW lr 3e-4, weight decay 1e-4, cosine annealing, 1000 epochs, batch 512, best-val checkpoint (val L1 every 100 epochs), seed 42 — the upstream `train_model` recipe.

Forward test MAE (A channel; C: mean of TE/TM) on the 50-sample test split defines the Δ -flag threshold (§4, k = 2).

3.4 Inverse design (the v8 core fix)

For each target spectrum $A^*(\lambda)$ (C: TE and TM jointly):

- **Variable:** $u \in [0,1]^d$ (normalized), mapped to physical units on the **design box** = dataset sampling box (`get_bounds`). Clamped to [0,1] after every step. This replaces v4a/v7's Adam-on-raw-nanometers, which could move at most ~22 nm in a 500 nm box (the frozen-optimizer bug).
- **Optimizer:** Adam on u , lr 0.05×500 iters, then 0.02×300 iters.
- **Multi-start:** R = 8 restarts per target — 1 mid-box + 7 uniform draws from `rng(1000 + target_idx)` — run as one batched tensor; the committed geometry is the restart with the lowest final surrogate loss. All restart endpoints are archived (they feed the Tier-1C dispersion diagnostic). (*Amendment, 2026-06-11, flagged by independent codex review: the as-run implementation seeds restarts with `rng(1000 + original-dataset index of the target)` — a fixed, pre-data-determined quantity, but not literally the wording above. All v8 results use the as-run definition; recorded here for exact reproducibility.*)
- **Objective:** mean squared error over the 100 wavelength channels between surrogate prediction and target (C: mean of TE and TM MSE).
- Physics features are recomputed differentially from the current geometry at every iteration; feature normalization uses the §3.3 TMM stats.

Targets: $N = 20$ per structure, drawn `rng(42).choice(test_split, 20, replace=False)` — random, from held-out data, identical mechanism for all three structures (this removes the curated-6-target confound of Structure A in manuscript v4).

3.5 Oracle revalidation

Every committed geometry is simulated with full-wave RCWA (`torcwa`) at the dataset-generation settings (64×64 grid, Fourier order [5,5]). **Convergence spot check** (pre-specified): the 3 highest- Δ geometries per structure are re-simulated at order [7,7]; the per-structure convergence floor is reported next to the Δ values it must not be confused with.

4. Metrics (pre-specified, unchanged thresholds)

- **Tier 1A — success rate**, $\tau = 5\%$: surrogate-claimed ($\text{Surr MAE} \leq \tau$), oracle ($\text{RCWA MAE} \leq \tau$). **Pretender** := $\text{Surr MAE} \leq \tau$ AND $\text{RCWA MAE} > \tau$ (formal definition, fixed before any v8 run). Wilson 95 % CIs on all counts including zeros.
- **Tier 1B — oracle gap**: $\Delta = \text{RCWA MAE} - \text{Surr MAE}$ at the committed geometry; flag when $\Delta > 2 \times$ forward-test-MAE. Report max Δ ; amplification (RCWA/Surr) only with the small-denominator caveat.
- **Tier 1C — convergence**: multi-start dispersion (std of the 8 final losses; max pairwise distance among the 8 endpoints in u-space) + loss-tail RSD of the committed run.
- **Taxonomy** (per committed geometry): T1 oracle divergence ($\text{RCWA MAE} > 3 \times \text{Surr MAE}$); T2 box-edge (any u within 0.05 of 0/1), against the d -dimensional uniform null; T3 mode collapse (NN distance $< 0.5 \times$ median pair distance), against the $N = 20$ uniform null (mean + 95 % band); **T4 surrogate-local smoothness — actually measured in v8** ($K = 100$ Gaussian perturbations, $\sigma = 1\%$ of box width, flag if mean perturbation $\text{MAE} > 5\%$).

5. Statistical analysis plan (corrects the v4 errors)

1. **Primary discrimination statistic**: $\rho(\text{RCWA MAE}, \text{Surr MAE})$ (Spearman, asymptotic p from `scipy`, approximate Bonett–Wright Fisher- z 95 % CI — no exact/permutation inference is claimed). This is the clean "does surrogate confidence track the reference-solver error" quantity.
2. $\rho(\Delta, \text{Surr MAE})$ is reported **only as secondary with the coupling caveat**: Δ contains $-\text{Surr}$ by construction, so its negative bias is partly tautological (in v4a data the coupling formula reproduced the observed -0.353 exactly — zero residual signal). No "confidence inversion" language unless the effect survives in $\rho(\text{RCWA}, \text{Surr})$ itself.
3. Cross-structure count comparisons (success, flags, pretenders): Fisher exact tests; we state outright when differences at $N = 20$ are not significant.
4. $N = 20$ with T3 clustering $\Rightarrow$ effective sample size may be smaller; the cluster structure (number of distinct modes, within-cluster spread) is reported alongside every count-based CI.
5. All seeds fixed and logged: split 42, train-subset 42, targets 42, restarts `rng(1000 + original-dataset index of the target)`, training 42 (multi-seed extension: train-subset/training seeds 123 and 777).

6. Corrections relative to manuscript v4 (why v8 numbers will differ)

Table S5: S4. Pre-specified protocol (verbatim) — protocol §6. Corrections relative to manuscript v4 (why v8 numbers will differ)

# v4/v7 defect	v8 fix
1 B/C run on 380–780 nm vs 400–1800 nm data (v4a)	native grids from dataset, pinned in artifacts
2 Adam on raw nm — optimizer frozen within ± 22 nm	normalized variables + multi-start (§3.4)
3 divergent RCWA rows in train/test/targets	Paper 1 filter (§3.1)
4 pretrain input column order permuted; phys stats recomputed on RCWA; C bounds mismatch	faithful loading (§3.3)
5 T4 reported but never computed	implemented (§4)
6 $\rho(\Delta, \text{Surr})$ headline (coupling artifact)	primary stat = $\rho(\text{RCWA}, \text{Surr})$ (§5)
7 A: 6 curated targets vs B: 20 random	identical $N = 20$ random targets for all structures
8 single deterministic init; <code>manual_seed</code> no-op	$R = 8$ multi-start, archived endpoints

7. Success criteria for "this is a paper"

The paper stands on whichever of these the v8 data delivers:

- **H1 confirmed (monotone degradation)**: the r diagnostic extends to inverse design → direct sequel to Paper 1.
- **H1 refuted (no relation / non-monotone)**: forward transferability does not certify inverse usability *and* the popular r-style diagnostic is shown insufficient — the protocol itself becomes the contribution, with the refutation as the empirical core.
- **Mixed**: the taxonomy localizes *which* failure mode tracks r and which does not — a structure-resolved reliability map.

What the paper can no longer claim regardless of outcome: the v4a confidence inversion, the 7.76× amplification, pretenders B-11/B-16, and any number produced by the frozen optimizer.

S5. Pilot-stage defects corrected before the protocol freeze

The earlier two/three-anchor pilots are superseded. The following defects, found in the full audit (docs/review_iter4_full_audit.md) and fixed here, are why the v8 numbers differ and why several earlier headlines are **withdrawn**:

Table S6: S5. Pilot-stage defects corrected before the protocol freeze

#	v4a/v7 defect	v8 fix
1	B/C inverse run on 380–780 nm against 400–1800 nm data	per-structure native grids, pinned in artifacts
2	Adam on raw nm — optimizer frozen within ±22 nm	normalized variables + multi-start (§2.2)
3	divergent RCWA rows leaked into train/test/targets	Paper 1 filter (§2.1)
4	permuted pretrain input columns; phys stats recomputed on RCWA; C-bounds mismatch	faithful strict loading (§2.1)
5	T4 reported but never computed	implemented and measured (§2.4)
6	$\rho(\Delta, \text{Surr})$ "confidence inversion" headline (coupling artifact)	primary stat = $\rho(\text{RCWA}, \text{Surr})$ (§2.5)
7	A used 6 curated targets vs B's 20 random	identical N = 20 random targets for all structures
8	single deterministic init; manual_seed no-op	R = 8 multi-start, archived endpoints

Consequently, the earlier **confidence inversion**, the **7.76× amplification**, and the named pretenders **B-11/B-16** are products of the frozen optimizer and the grid mismatch and are **retracted**. No claim in this manuscript depends on them.

S6. Per-design table

Source: results_pub/ (reliability, recoverability JSONs; rcwa npz). $\tau = 5\%$. status: pretender = $\text{Surr} \leq \tau$ & $\text{RCWA} > \tau$; confirmed = both $\leq \tau$; honest-failure = $\text{Surr} > \tau$; degenerate = solver failure. support viol.: the committed geometry lies outside the physical support of the parametrization (a patch width $W \geq P$ for A and C, $R_{\text{in}} \geq R_{\text{out}}$ for B), which is a narrower test than the generator-feasibility rule of Section 3.8.

Table S7: S6. Per-design table

S	seed	#	idx	Surr MAE %	RCWA MAE %	Δ pp	status	T1	T2	T3	T4	Surr@truth %	$\ u^* - u_{\text{true}}\ $	support viol.	peak shift nm	peak ΔA
A	42	1	182	0.75	3.20	+2.45	confirmed	1	1	0	0	2.09	1.25	0	+1046	+0.04
A	42	2	366	0.80	4.38	+3.58	confirmed	1	1	0	0	4.32	0.62	1	-14	+0.00
A	42	3	109	0.47	3.23	+2.76	confirmed	1	1	0	0	1.69	1.04	1	+99	+0.02
A	42	4	437	0.56	2.96	+2.40	confirmed	1	0	0	0	1.72	1.15	0	-127	+0.00
A	42	5	462	0.33	3.53	+3.20	confirmed	1	1	0	0	1.02	1.00	1	+57	-0.01
A	42	6	22	0.61	1.67	+1.06	confirmed	0	1	0	0	1.85	0.53	0	-71	+0.00
A	42	7	316	0.80	1.39	+0.59	confirmed	0	1	0	0	1.69	0.71	0	+1145	-0.03
A	42	8	420	0.16	0.63	+0.46	confirmed	1	1	0	0	1.15	0.85	1	+0	+0.01
A	42	9	459	0.48	3.28	+2.80	confirmed	1	0	0	0	2.15	0.62	1	+0	+0.00
A	42	10	97	0.93	2.04	+1.11	confirmed	0	1	0	0	1.03	0.86	0	-820	+0.01
A	42	11	110	0.31	1.43	+1.12	confirmed	1	1	0	0	0.73	0.46	0	-28	+0.01
A	42	12	272	0.15	1.01	+0.86	confirmed	1	1	0	0	0.64	0.93	0	-28	+0.01
A	42	13	35	0.23	10.13	+9.90	pretender	1	1	0	0	0.76	0.87	1	+0	+0.24
A	42	14	481	0.12	1.45	+1.33	confirmed	1	0	0	0	1.37	0.41	0	+28	-0.01
A	42	15	237	0.73	2.75	+2.02	confirmed	1	1	0	0	3.21	1.04	1	+14	+0.11
A	42	16	41	0.30	1.42	+1.12	confirmed	1	1	0	0	1.26	0.50	0	-42	-0.00

S	seed	#	idx	Surr MAE %	RCWA MAE %	Δ pp	status	T1	T2	T3	T4	Surr@truth %	$\ u^*-u_{true}\ $	support viol.	peak shift nm	peak ΔA
A	42	17	320	0.20	1.23	+1.03	confirmed	1	0	0	0	1.51	0.54	1	+85	+0.05
A	42	18	240	0.79	1.50	+0.71	confirmed	0	0	0	0	2.14	0.57	0	+0	-0.00
A	42	19	6	0.11	1.68	+1.56	confirmed	1	1	0	0	3.56	0.98	1	+0	-0.02
A	42	20	396	0.34	2.46	+2.12	confirmed	1	0	0	0	2.32	0.69	0	+0	+0.01
A	123	1	182	0.58	2.77	+2.19	confirmed	1	1	0	0	1.49	1.29	0	+1202	-0.01
A	123	2	366	0.90	4.70	+3.80	confirmed	1	1	0	0	3.05	1.13	1	+28	-0.07
A	123	3	109	0.45	4.43	+3.98	confirmed	1	1	1	0	1.96	0.95	1	-240	+0.02
A	123	4	437	0.76	4.86	+4.11	confirmed	1	1	0	0	2.95	0.56	1	+57	-0.13
A	123	5	462	0.35	1.90	+1.55	confirmed	1	1	0	0	1.55	0.50	1	+57	+0.05
A	123	6	22	0.35	1.82	+1.47	confirmed	1	1	0	0	1.90	0.28	0	+14	+0.01
A	123	7	316	0.71	2.14	+1.43	confirmed	1	1	0	0	1.94	0.61	0	+42	+0.06
A	123	8	420	0.11	0.83	+0.72	confirmed	1	0	0	0	1.00	0.44	0	+0	+0.01
A	123	9	459	0.84	1.88	+1.03	confirmed	0	1	0	0	2.11	0.60	0	+0	-0.01
A	123	10	97	0.47	1.38	+0.91	confirmed	0	1	0	0	1.32	0.52	0	+0	+0.02
A	123	11	110	0.29	2.72	+2.43	confirmed	1	1	1	0	1.21	0.40	1	-14	+0.02
A	123	12	272	0.13	1.58	+1.45	confirmed	1	1	0	0	0.77	0.89	1	-28	-0.01
A	123	13	35	0.20	3.53	+3.33	confirmed	1	1	0	0	1.20	1.21	1	+28	+0.04
A	123	14	481	0.06	0.58	+0.52	confirmed	1	1	0	0	0.88	0.71	0	+0	-0.00
A	123	15	237	1.28	2.95	+1.68	confirmed	0	1	0	0	2.43	0.82	0	-57	+0.05
A	123	16	41	0.54	0.91	+0.37	confirmed	0	1	0	0	0.85	0.72	0	+28	-0.01
A	123	17	320	0.17	0.91	+0.74	confirmed	1	1	0	0	1.59	0.78	0	+0	+0.01
A	123	18	240	1.04	1.76	+0.72	confirmed	0	0	0	0	3.30	0.61	0	+0	-0.01
A	123	19	6	0.12	2.43	+2.31	confirmed	1	1	0	0	3.89	0.55	0	+0	-0.05
A	123	20	396	0.34	2.65	+2.30	confirmed	1	1	0	0	1.40	1.02	0	+0	+0.01
A	777	1	182	0.77	1.76	+0.99	confirmed	0	1	0	0	2.61	0.97	0	+1174	-0.01
A	777	2	366	0.53	2.67	+2.14	confirmed	1	1	0	0	3.48	0.37	0	+42	-0.02
A	777	3	109	0.47	1.39	+0.91	confirmed	0	1	0	0	1.80	0.54	1	+28	-0.00
A	777	4	437	0.66	3.80	+3.13	confirmed	1	1	0	0	1.62	0.73	0	-99	+0.01
A	777	5	462	0.35	2.72	+2.37	confirmed	1	1	0	0	1.39	1.55	1	+57	+0.00
A	777	6	22	1.07	1.59	+0.51	confirmed	0	1	0	0	1.59	0.99	0	+28	+0.02
A	777	7	316	0.84	1.42	+0.58	confirmed	0	0	0	0	1.39	0.67	0	-28	-0.03
A	777	8	420	0.15	0.98	+0.82	confirmed	1	0	0	0	1.21	0.58	0	+0	-0.01
A	777	9	459	0.83	3.16	+2.34	confirmed	1	1	0	0	2.43	0.41	0	-1400	+0.02
A	777	10	97	0.50	1.67	+1.18	confirmed	1	1	0	0	1.58	0.68	1	+0	+0.02
A	777	11	110	0.43	—	—	degenerate	0	1	0	0	1.22	0.43	1	—	—
A	777	12	272	0.09	0.55	+0.45	confirmed	1	0	0	0	1.17	0.72	0	-14	+0.00
A	777	13	35	0.23	3.98	+3.74	confirmed	1	1	0	0	0.89	1.27	1	-85	-0.09
A	777	14	481	0.15	1.27	+1.12	confirmed	1	0	0	0	0.87	0.50	0	+0	+0.01
A	777	15	237	0.55	2.96	+2.41	confirmed	1	1	0	0	2.07	0.91	0	+0	+0.12
A	777	16	41	0.60	1.03	+0.43	confirmed	0	1	0	0	1.03	0.65	0	-28	-0.01
A	777	17	320	0.12	2.42	+2.30	confirmed	1	1	0	0	1.52	0.78	0	+156	+0.06
A	777	18	240	0.89	1.83	+0.94	confirmed	0	0	0	0	2.96	0.55	0	+0	-0.00
A	777	19	6	0.22	5.32	+5.10	pretender	1	1	0	0	2.96	0.68	1	+0	+0.09
A	777	20	396	0.17	1.73	+1.55	confirmed	1	0	0	0	1.33	0.41	0	+0	-0.03
B	42	1	182	0.29	4.79	+4.50	confirmed	1	1	0	0	0.85	0.54	1	+0	-0.07
B	42	2	366	0.39	2.38	+2.00	confirmed	1	1	0	0	2.28	1.06	0	+113	+0.06
B	42	3	109	0.35	2.77	+2.41	confirmed	1	1	0	0	1.98	0.76	0	-28	-0.03
B	42	4	438	0.44	2.24	+1.80	confirmed	1	1	0	0	1.44	0.72	0	-42	+0.01
B	42	5	462	1.05	4.82	+3.77	confirmed	1	1	1	0	2.00	0.77	0	-42	-0.04
B	42	6	22	0.54	3.89	+3.35	confirmed	1	1	0	0	1.70	0.71	0	-1145	-0.03
B	42	7	316	0.12	4.15	+4.04	confirmed	1	0	1	0	1.10	0.11	0	-57	+0.00
B	42	8	421	0.95	2.88	+1.93	confirmed	1	1	0	0	2.94	0.53	0	+71	-0.03
B	42	9	459	1.19	3.46	+2.27	confirmed	0	0	0	0	2.80	0.74	0	+354	-0.02
B	42	10	97	0.46	2.14	+1.68	confirmed	1	1	0	0	2.07	0.34	0	-28	+0.01
B	42	11	110	1.27	1.21	-0.06	confirmed	0	1	0	0	1.31	0.80	0	-28	+0.02
B	42	12	272	0.38	12.49	+12.11	pretender	1	1	0	0	2.31	1.26	0	-99	-0.08
B	42	13	35	0.23	2.94	+2.71	confirmed	1	0	1	0	1.47	0.30	0	+14	+0.01
B	42	14	480	0.11	1.25	+1.14	confirmed	1	0	1	0	0.41	0.27	0	+0	+0.00
B	42	15	237	0.24	1.49	+1.25	confirmed	1	0	1	0	0.83	0.31	0	+0	+0.00
B	42	16	41	0.40	6.06	+5.65	pretender	1	1	0	0	1.53	0.58	0	+0	-0.03
B	42	17	320	0.78	4.48	+3.70	confirmed	1	1	0	0	1.94	0.47	0	-14	+0.04
B	42	18	240	0.18	1.96	+1.77	confirmed	1	1	1	0	1.82	0.19	0	-28	-0.00
B	42	19	6	0.26	8.35	+8.09	pretender	1	1	1	0	1.44	1.03	1	+156	+0.10
B	42	20	397	0.33	0.97	+0.65	confirmed	0	1	0	0	1.28	0.88	0	-28	-0.00
B	123	1	182	0.23	1.37	+1.15	confirmed	1	0	0	0	0.73	0.26	0	+0	-0.01
B	123	2	366	0.56	2.66	+2.11	confirmed	1	1	1	0	1.47	0.91	0	+127	+0.04
B	123	3	109	0.34	1.39	+1.05	confirmed	1	1	0	0	1.55	0.97	0	-28	-0.01
B	123	4	438	0.43	0.97	+0.54	confirmed	0	0	1	0	1.41	0.63	0	-28	-0.00
B	123	5	462	1.27	6.41	+5.14	pretender	1	1	1	0	1.40	0.50	0	-42	+0.02
B	123	6	22	0.57	1.69	+1.12	confirmed	0	1	1	0	1.60	0.70	0	+0	+0.00
B	123	7	316	0.22	4.17	+3.95	confirmed	1	0	1	0	0.79	0.57	0	-71	+0.00
B	123	8	421	0.92	1.67	+0.75	confirmed	0	1	0	0	2.58	0.51	0	+28	-0.00
B	123	9	459	1.30	2.61	+1.31	confirmed	0	1	0	0	2.44	0.47	0	+0	-0.03
B	123	10	97	0.69	3.80	+3.12	confirmed	1	1	0	0	2.96	0.50	0	+85	-0.01
B	123	11	110	0.93	2.08	+1.15	confirmed	0	1	0	0	0.81	0.94	0	-42	+0.01
B	123	12	272	0.21	2.04	+1.84	confirmed	1	1	1	0	2.47	1.03	0	-170	+0.04
B	123	13	35	0.40	1.75	+1.35	confirmed	1	0	1	0	1.73	0.50	0	+0	+0.00
B	123	14	480	0.59	1.64	+1.05	confirmed	0	1	0	0	0.41	1.00	0	-71	+0.02
B	123	15	237	0.24	0.96	+0.73	confirmed	1	0	1	0	0.52	0.33	0	+14	-0.01
B	123	16	41	0.41	1.05	+0.64	confirmed	0	0	1	0	1.05	0.61	0	+28	+0.00
B	123	17	320	0.74	2.54	+1.80	confirmed	1	1	0	0	1.48	0.76	1	+28	-0.03
B	123	18	240	0.33	5.22	+4.89	pretender	1	1	1	0	1.92	0.45	0	-99	+0.06
B	123	19	6	0.25	15.92	+15.67	pretender	1	1	1	0	1.25	1.14	1	+42	+0.20
B	123	20	397	0.67	1.06	+0.39	confirmed	0	1	0	0	1.42	0.93	0	-42	-0.00
B	777	1	182	0.24	6.98	+6.74	pretender	1	1	1	0	0.92	0.95	1	-99	+0.07

S	seed	#	idx	Surr MAE %	RCWA MAE %	Δ pp	status	T1	T2	T3	T4	Surr@truth %	$\ u^*-u_{\text{true}}\ $	support viol.	peak shift nm	peak ΔA
B	777	2	366	0.49	2.14	+1.65	confirmed	1	1	0	0	1.62	0.35	0	+99	+0.02
B	777	3	109	0.33	0.94	+0.62	confirmed	0	1	0	0	1.45	0.82	0	+42	-0.00
B	777	4	438	0.57	0.76	+0.20	confirmed	0	1	1	0	1.47	0.54	0	-57	-0.00
B	777	5	462	1.01	4.71	+3.70	confirmed	1	0	1	0	1.59	0.24	0	-42	+0.04
B	777	6	22	0.40	1.60	+1.20	confirmed	1	1	0	0	1.26	0.47	0	+0	+0.00
B	777	7	316	0.13	4.00	+3.87	confirmed	1	0	1	0	0.77	0.45	0	-57	-0.00
B	777	8	421	1.00	2.40	+1.40	confirmed	0	1	0	0	2.01	0.52	0	+57	-0.02
B	777	9	459	1.39	3.46	+2.06	confirmed	0	0	0	0	2.59	0.70	0	+170	-0.02
B	777	10	97	0.86	7.34	+6.48	pretender	1	1	0	0	3.29	1.27	0	+212	+0.08
B	777	11	110	1.03	2.46	+1.43	confirmed	0	1	0	0	1.14	1.00	0	+99	-0.02
B	777	12	272	0.26	2.89	+2.63	confirmed	1	1	0	0	2.69	0.96	0	-113	+0.02
B	777	13	35	0.42	1.59	+1.17	confirmed	1	1	1	0	1.82	0.47	0	-71	-0.01
B	777	14	480	0.18	2.25	+2.07	confirmed	1	0	0	0	0.41	0.36	0	-28	-0.01
B	777	15	237	0.48	1.17	+0.70	confirmed	0	0	0	0	0.81	0.38	0	+14	-0.01
B	777	16	41	0.73	0.79	+0.06	confirmed	0	0	1	0	1.11	0.25	0	+99	-0.00
B	777	17	320	0.40	1.32	+0.92	confirmed	1	1	0	0	1.53	0.44	0	+14	+0.03
B	777	18	240	0.09	7.73	+7.63	pretender	1	0	0	0	2.43	0.43	0	-141	-0.03
B	777	19	6	0.32	2.33	+2.00	confirmed	1	0	1	0	0.97	0.54	1	-14	-0.01
B	777	20	397	0.44	1.18	+0.74	confirmed	0	1	0	0	1.00	0.38	0	+0	+0.00
C	42	1	187	1.58	1.52	-0.06	confirmed	0	1	0	0	2.09	0.89	0	+14/+0	-0.00/-0.02
C	42	2	376	0.96	2.27	+1.31	confirmed	0	1	0	0	1.60	0.51	0	+0/-57	+0.00/+0.04
C	42	3	325	1.05	1.51	+0.46	confirmed	0	1	0	0	1.84	0.21	0	-28/-57	+0.02/-0.00
C	42	4	449	1.14	0.95	-0.19	confirmed	0	0	1	0	1.23	0.17	0	+0/+85	-0.01/+0.00
C	42	5	474	0.31	0.31	+0.00	confirmed	0	0	1	0	0.38	0.18	0	+0/+0	-0.00/-0.00
C	42	6	24	0.76	1.16	+0.39	confirmed	0	1	0	0	1.59	0.37	0	+0/-226	-0.01/-0.01
C	42	7	443	2.58	3.86	+1.27	confirmed	0	1	0	0	3.96	0.30	0	-28/-14	-0.02/-0.02
C	42	8	25	2.73	1.55	-1.18	confirmed	0	1	0	0	3.09	0.11	0	+28/+14	-0.02/+0.01
C	42	9	471	0.28	0.32	+0.04	confirmed	0	0	1	0	0.46	0.70	0	+42/-57	-0.00/-0.00
C	42	10	99	0.78	1.27	+0.49	confirmed	0	1	0	0	2.66	0.95	0	-42/+113	-0.01/+0.03
C	42	11	112	1.13	1.08	-0.05	confirmed	0	0	1	0	1.86	0.63	0	+42/+14	-0.00/+0.00
C	42	12	431	1.39	0.89	-0.50	confirmed	0	0	1	0	1.68	0.33	0	+57/+0	-0.01/-0.00
C	42	13	37	1.25	1.41	+0.16	confirmed	0	1	0	0	1.92	0.13	0	+14/-14	+0.00/-0.01
C	42	14	481	1.15	2.26	+1.11	confirmed	0	0	0	0	1.65	0.16	0	-679/+57	+0.00/+0.01
C	42	15	244	1.57	1.66	+0.09	confirmed	0	0	0	0	2.18	0.33	0	+28/+28	-0.00/-0.00
C	42	16	43	0.16	0.42	+0.26	confirmed	0	0	0	0	0.28	0.08	0	+28/+14	-0.00/-0.00
C	42	17	329	0.14	0.16	+0.02	confirmed	0	0	0	0	0.26	0.11	0	+0/+0	+0.00/+0.00
C	42	18	247	3.85	5.93	+2.08	pretender	0	0	0	0	5.16	0.31	0	+14/+28	-0.00/-0.07
C	42	19	6	1.36	1.67	+0.31	confirmed	0	0	0	0	1.62	0.16	0	+28/+28	-0.01/-0.00
C	42	20	406	2.07	6.73	+4.66	pretender	1	1	0	0	6.30	0.74	0	-240/+71	-0.04/+0.07
C	123	1	187	1.02	0.92	-0.10	confirmed	0	1	0	0	2.26	0.67	0	+28/+0	-0.00/-0.00
C	123	2	376	0.82	1.84	+1.02	confirmed	0	1	1	0	1.27	0.51	0	+14/+141	+0.00/+0.01
C	123	3	325	1.01	1.20	+0.19	confirmed	0	1	1	0	1.52	0.16	0	+28/+0	-0.01/-0.03
C	123	4	449	0.94	1.61	+0.68	confirmed	0	0	1	0	1.57	0.28	0	-28/+14	-0.03/-0.00
C	123	5	474	0.34	0.36	+0.02	confirmed	0	0	1	0	0.47	0.38	0	+0/+0	-0.00/-0.00
C	123	6	24	0.96	1.47	+0.52	confirmed	0	0	0	0	1.33	0.30	0	+28/-198	-0.02/+0.00
C	123	7	443	1.69	2.95	+1.26	confirmed	0	1	0	0	4.06	0.77	0	-28/+28	+0.04/-0.01
C	123	8	25	3.10	4.37	+1.28	confirmed	0	1	0	0	3.13	0.23	0	-113/+141	-0.02/-0.01
C	123	9	471	0.24	0.53	+0.29	confirmed	0	1	1	0	0.51	0.91	0	+14/+14	-0.00/-0.01
C	123	10	99	0.50	1.70	+1.20	confirmed	1	1	0	0	2.15	1.15	0	+14/-57	+0.01/+0.05
C	123	11	112	1.00	1.05	+0.05	confirmed	0	0	1	0	1.74	0.62	0	+28/+14	-0.00/+0.00
C	123	12	431	1.19	1.13	-0.06	confirmed	0	0	1	0	1.87	0.24	0	+71/+14	-0.01/-0.01
C	123	13	37	1.39	2.05	+0.65	confirmed	0	0	0	0	1.65	0.36	0	+42/+0	-0.02/-0.01
C	123	14	481	1.30	1.17	-0.13	confirmed	0	0	0	0	1.34	0.14	0	-721/+28	+0.00/+0.01
C	123	15	244	1.43	1.93	+0.50	confirmed	0	1	0	0	1.95	0.13	0	+57/+28	+0.02/-0.01
C	123	16	43	0.14	0.26	+0.11	confirmed	0	0	1	0	0.22	0.06	0	+14/+14	-0.00/-0.00
C	123	17	329	0.17	0.12	-0.05	confirmed	0	0	1	0	0.25	0.42	0	+0/+0	+0.00/+0.00
C	123	18	247	4.63	7.75	+3.11	pretender	0	1	0	0	5.15	0.34	0	-42/+14	-0.09/-0.02
C	123	19	6	1.43	1.86	+0.42	confirmed	0	1	0	0	1.50	0.16	0	-14/-14	-0.01/+0.00
C	123	20	406	1.99	4.55	+2.56	confirmed	0	1	0	0	4.65	0.73	0	-57/+467	-0.01/+0.01
C	777	1	187	0.91	1.68	+0.76	confirmed	0	1	0	0	2.81	0.83	0	+14/+0	-0.00/-0.02
C	777	2	376	0.90	1.61	+0.72	confirmed	0	1	1	0	1.42	0.21	0	-14/+127	+0.00/-0.00
C	777	3	325	0.76	1.51	+0.75	confirmed	0	1	1	0	1.25	0.26	0	+42/+14	-0.01/+0.02
C	777	4	449	1.00	0.71	-0.30	confirmed	0	1	0	0	1.32	0.37	0	+0/+42	+0.02/+0.01
C	777	5	474	0.32	0.24	-0.08	confirmed	0	0	0	0	0.44	0.33	0	+0/+0	-0.00/-0.00
C	777	6	24	0.86	2.48	+1.62	confirmed	0	1	0	0	1.51	0.53	0	+0/-14	+0.02/-0.00
C	777	7	443	1.47	1.99	+0.53	confirmed	0	1	0	0	2.37	0.34	0	-57/-14	+0.01/-0.01
C	777	8	25	2.68	2.62	-0.05	confirmed	0	1	1	0	2.66	0.23	0	-14/-42	-0.01/-0.00
C	777	9	471	0.25	0.57	+0.33	confirmed	0	0	1	0	0.54	0.56	0	+28/+14	-0.00/-0.01
C	777	10	99	1.25	1.86	+0.60	confirmed	0	1	0	0	1.86	0.89	0	+14/-113	-0.00/-0.02
C	777	11	112	1.15	1.02	-0.14	confirmed	0	0	0	0	1.78	0.65	0	+28/+14	-0.00/+0.01
C	777	12	431	1.16	1.41	+0.25	confirmed	0	0	1	0	1.55	0.33	0	+0/+28	-0.02/-0.01
C	777	13	37	1.03	0.89	-0.15	confirmed	0	1	0	0	1.66	0.17	0	-28/+0	-0.01/-0.00
C	777	14	481	1.57	1.28	-0.29	confirmed	0	0	0	0	1.99	0.14	0	+0/-14	-0.00/-0.01
C	777	15	244	1.34	2.22	+0.88	confirmed	0	0	0	0	2.67	0.34	0	+0/+14	+0.02/+0.01
C	777	16	43	0.17	0.25	+0.08	confirmed	0	0	0	0	0.24	0.10	0	+14/+14	-0.00/-0.00
C	777	17	329	0.18	0.16	-0.02	confirmed	0	0	0	0	0.22	0.13	0	+0/+0	+0.00/-0.00
C	777	18	247	3.72	6.13	+2.41	pretender	0	0	1	0	4.38	0.11	0	+28/+28	-0.01/-0.06
C	777	19	6	1.27	1.42	+0.15	confirmed	0	0	0	0	1.55	0.16	0	+0/+0	-0.02/-0.00
C	777	20	406	3.36	5.24	+1.88	pretender	0	0	0	0	5.45	0.78	0	+0/+57	+0.01/+0.06

Valid designs: 179; pretenders: 16.

The machine-readable version, with the committed and true geometry of every design, the smallest

feature, the pixel size, the generator rules each design violates and the peak-wavelength shift, is `table_S6_pretenders_v8.csv`, available from the corresponding author (planned deposit).

S7. Supplementary figures

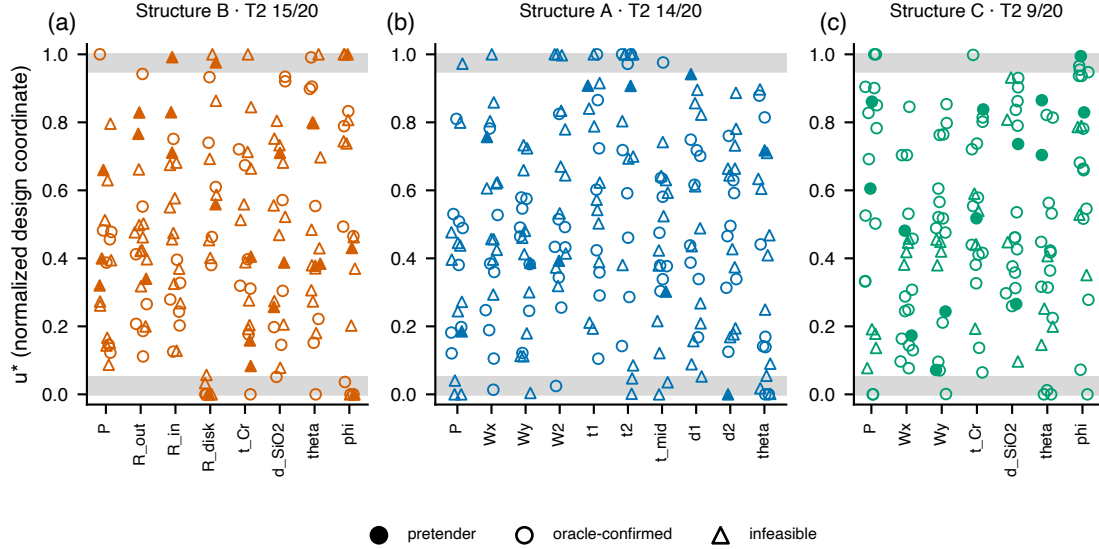


Figure S1: Per-parameter position of every committed geometry inside the design box (seed 42; one column per design parameter, y = the normalized coordinate u^*). Grey bands mark the T2 box-edge region, within 0.05 of either bound. Filled markers are pretenders and open markers oracle-confirmed designs; triangles are geometries that violate a generator constraint. The in-band counts equal the T2 totals of Supplementary Table S1.

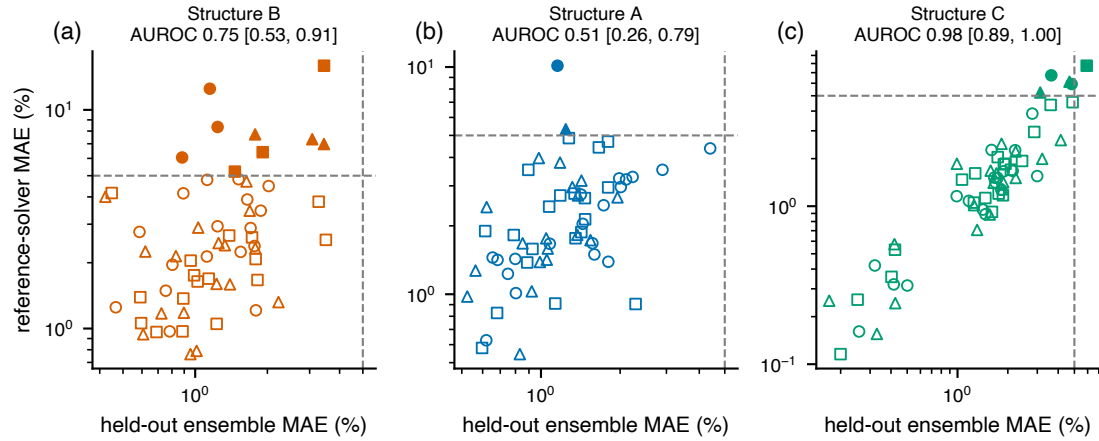


Figure S2: Held-out ensemble error against reference-solver error for every valid committed geometry, per structure and training seed (log-log). The held-out ensemble error is the mean absolute error of the two surrogates that did *not* commit the design, so it is available without any solver call. Dashed lines mark $\tau = 5\%$; filled markers are pretenders. Panel titles give the pooled AUROC for separating pretenders from oracle-confirmed designs, with its bootstrap 95% interval.

S8. The as-submitted arm of Section 3.10

The identical protocol run end to end on the release of [1] that the published pipeline superseded (upstream commit 920b1bd; fixed Fourier order 5, complex128; datasets struct_A_vis_500 (479 good, 380–780 nm), struct_B_500 (461, 400–1800 nm), struct_C_500 (400, 400–1800 nm); C trained on 300 samples with a [50, 400] nm normalization box and 13 physics features). Artifacts: results_v8/. The tables below are the S1, S2 and S6 equivalents for that arm.

S8.1. Per-run reliability accounting (as-submitted arm)

Table S8: S8. The as-submitted arm of Section 3.10

S	seed	n_valid	surrogate-claimed [Wilson 95 %]	oracle-confirmed [Wilson]	pretenders [Wilson]	Δ -flagged [Wilson]	flag thr (%)	modes	T1	T2	T3	T4	tail RSD	endpoint spread (u)	ρ (RCWA, Surr) [CI95]	p
A	42	20	20 [83.9, 100.0]	11 [34.2, 74.2]	9 [25.8, 65.8]	0 [0.0, 16.1]	9.69	20	19	15	0	0	0.0175	1.978	+0.18 [-0.28, 0.58]	0.435
A	123	20	20 [83.9, 100.0]	11 [34.2, 74.2]	9 [25.8, 65.8]	1 [0.9, 23.6]	9.68	18	19	16	3	0	0.0257	1.98	+0.39 [-0.08, 0.72]	0.086
A	777	20	20 [83.9, 100.0]	11 [34.2, 74.2]	9 [25.8, 65.8]	1 [0.9, 23.6]	9.46	19	20	16	2	0	0.0138	2.019	+0.45 [-0.02, 0.75]	0.048
B	42	20	20 [83.9, 100.0]	11 [34.2, 74.2]	9 [25.8, 65.8]	6 [14.5, 51.9]	6.37	17	16	14	5	0	0.0127	1.64	-0.03 [-0.47, 0.42]	0.890
B	123	20	20 [83.9, 100.0]	9 [25.8, 65.8]	11 [34.2, 74.2]	8 [21.9, 61.3]	6.52	18	20	16	4	0	0.0084	1.643	-0.11 [-0.53, 0.35]	0.654
B	777	20	20 [83.9, 100.0]	10 [29.9, 70.1]	10 [29.9, 70.1]	6 [14.5, 51.9]	6.56	15	18	14	7	0	0.0239	1.69	+0.20 [-0.27, 0.60]	0.391
C	42	20	20 [83.9, 100.0]	7 [18.1, 56.7]	13 [43.3, 81.9]	3 [5.2, 36.0]	7.73	15	11	11	7	0	0.0171	1.62	-0.04 [-0.47, 0.41]	0.880
C	123	20	20 [83.9, 100.0]	8 [21.9, 61.3]	12 [38.7, 78.1]	2 [2.8, 30.1]	7.66	13	11	12	10	0	0.0145	1.624	+0.53 [0.08, 0.80]	0.016
C	777	19	19 [83.2, 100.0]	6 [15.4, 54.0]	13 [46.0, 84.6]	2 [2.9, 31.4]	8.0	15	11	12	7	0	0.0021	1.676	+0.28 [-0.21, 0.65]	0.254
all		179			95/179 (A 27/60, B 30/60, C 38/59)											

S8.2. Commitment-rule comparison (seed 42; as-submitted arm)

Table S9: S8. The as-submitted arm of Section 3.10

S	rule	surrogate-claimed	oracle-confirmed	pretenders (of n)	pretenders / claimed [Wilson]	mean oracle MAE (%)	Fisher p vs best-of-8 (design-level)	Fisher p (conditional on claimed)
A	gradient best-of-8	20	11	9/20	9/20 [25.8, 65.8]	4.85	—	—
A	random search (budget-matched)	19	9	10/20	10/19 [31.7, 72.7]	5.45	1.000	0.752
A	single start (mid-box)	20	12	8/20	8/20 [21.9, 61.3]	5.49	1.000	1.000
B	gradient best-of-8	20	11	9/20	9/20 [25.8, 65.8]	6.65	—	—
B	random search (budget-matched)	19	5	14/20	14/19 [51.2, 88.2]	10.79	0.200	0.105
B	single start (mid-box)	20	11	9/20	9/20 [25.8, 65.8]	5.96	1.000	1.000
C	gradient best-of-8	20	7	13/20	13/20 [43.3, 81.9]	6.85	—	—
C	random search (budget-matched)	16	0	16/20	16/16 [80.6, 100.0]	9.17	0.480	0.011
C	single start (mid-box)	17	8	9/20	9/17 [31.0, 73.8]	7.18	0.341	0.516

S8.3. Per-design table (as-submitted arm)

Source: results_v8/ (reliability, recoverability JSONs; rcwa npz). $\tau = 5\%$. status: pretender = $\text{Surr} \leq \tau$ & $\text{RCWA} > \tau$; confirmed = both $\leq \tau$; honest-failure = $\text{Surr} > \tau$; degenerate = solver failure. support viol. as in Section S6.

Table S10: S8. The as-submitted arm of Section 3.10

S	seed	#	idx	Surr MAE %	RCWA MAE %	Δ pp	status	T1	T2	T3	T4	Surr@truth %	$\ u^* - u_{\text{true}}\ $	support viol.	peak shift nm	peak ΔA
A	42	1	190	0.07	1.74	+1.67	confirmed	1	1	0	0	2.73	1.15	0	+0	-0.01
A	42	2	381	0.56	2.84	+2.28	confirmed	1	1	0	0	4.01	1.15	0	-12	+0.02
A	42	3	451	1.22	2.74	+1.52	confirmed	0	1	0	0	5.13	1.01	0	+12	-0.01
A	42	4	456	0.05	2.54	+2.49	confirmed	1	1	0	0	0.66	1.47	1	+0	+0.00
A	42	5	482	0.17	7.56	+7.38	pretender	1	1	0	0	2.72	1.12	1	+109	+0.11
A	42	6	24	0.47	7.42	+6.95	pretender	1	1	0	0	5.83	0.75	0	-48	+0.18
A	42	7	16	0.47	3.26	+2.79	confirmed	1	1	0	0	9.79	1.43	1	+4	+0.10
A	42	8	289	0.27	2.90	+2.64	confirmed	1	0	0	0	2.65	1.24	0	+4	+0.02
A	42	9	327	0.21	5.83	+5.62	pretender	1	1	0	0	6.77	1.41	1	-28	+0.01
A	42	10	101	0.63	6.76	+6.13	pretender	1	1	0	0	4.37	1.66	0	+323	+0.03
A	42	11	249	1.24	4.44	+3.19	confirmed	1	1	0	0	7.54	1.11	0	-24	-0.01
A	42	12	25	0.45	3.06	+2.61	confirmed	1	0	0	0	5.14	0.74	0	+12	+0.08
A	42	13	37	1.00	4.26	+3.26	confirmed	1	0	0	0	4.21	1.36	1	-16	-0.02
A	42	14	481	0.21	6.50	+6.29	pretender	1	1	0	0	6.22	1.18	1	+61	+0.12
A	42	15	246	1.43	6.48	+5.05	pretender	1	1	0	0	5.36	0.78	0	+20	-0.02
A	42	16	43	0.37	6.73	+6.36	pretender	1	1	0	0	3.20	0.99	1	+28	+0.12
A	42	17	333	0.60	4.22	+3.63	confirmed	1	0	0	0	5.10	1.08	0	+20	+0.06
A	42	18	312	1.46	7.36	+5.90	pretender	1	0	0	0	6.63	0.74	1	-16	+0.15
A	42	19	6	0.20	5.92	+5.72	pretender	1	1	0	0	4.44	0.92	1	-4	+0.12
A	42	20	413	0.08	4.36	+4.28	confirmed	1	1	0	0	0.96	1.39	1	+0	+0.04
A	123	1	190	0.04	2.14	+2.10	confirmed	1	1	0	0	3.85	0.99	0	+0	-0.01
A	123	2	381	0.73	4.04	+3.31	confirmed	1	1	0	0	4.83	1.45	0	+0	+0.05
A	123	3	451	1.04	4.60	+3.56	confirmed	1	1	0	0	5.79	0.73	1	+12	+0.02
A	123	4	456	0.06	10.08	+10.02	pretender	1	1	0	0	0.61	1.33	1	+77	+0.31
A	123	5	482	0.10	4.37	+4.27	confirmed	1	1	0	0	3.12	1.45	1	+0	-0.03
A	123	6	24	0.31	8.76	+8.45	pretender	1	1	0	0	6.51	1.10	0	-81	+0.14
A	123	7	16	0.50	7.13	+6.63	pretender	1	1	0	0	9.40	1.65	0	-97	-0.01
A	123	8	289	0.32	3.97	+3.65	confirmed	1	1	1	0	3.49	1.07	0	-8	+0.03
A	123	9	327	0.52	5.81	+5.29	pretender	1	1	0	0	7.09	1.39	0	-28	-0.08
A	123	10	101	0.53	3.98	+3.45	confirmed	1	1	0	0	2.21	1.61	1	+8	+0.11
A	123	11	249	0.57	4.65	+4.08	confirmed	1	0	0	0	5.47	0.56	0	-8	-0.01
A	123	12	25	0.39	9.12	+8.73	pretender	1	0	1	0	4.30	1.17	0	-20	+0.04
A	123	13	37	0.75	7.94	+7.19	pretender	1	0	0	0	4.96	1.38	1	+12	+0.06
A	123	14	481	0.25	3.50	+3.25	confirmed	1	1	0	0	6.43	0.95	1	-12	+0.03
A	123	15	246	2.63	6.81	+4.18	pretender	0	1	0	0	4.44	0.50	0	+32	+0.05
A	123	16	43	0.24	2.82	+2.59	confirmed	1	1	0	0	3.34	0.90	0	-8	+0.03
A	123	17	333	0.65	8.08	+7.42	pretender	1	0	1	0	3.18	0.96	0	+36	-0.02
A	123	18	312	2.33	9.62	+7.29	pretender	1	1	0	0	8.95	0.74	1	-24	+0.13
A	123	19	6	0.11	3.17	+3.06	confirmed	1	1	0	0	10.10	0.99	0	-32	-0.00
A	123	20	413	0.06	4.26	+4.20	confirmed	1	1	0	0	0.54	1.52	0	+0	-0.05
A	777	1	190	0.10	1.65	+1.55	confirmed	1	0	1	0	1.36	1.49	1	+0	-0.01
A	777	2	381	0.64	4.80	+4.16	confirmed	1	1	0	0	2.92	1.08	1	+0	+0.02
A	777	3	451	0.79	4.04	+3.25	confirmed	1	0	0	0	5.63	1.00	0	+12	-0.00
A	777	4	456	0.05	3.82	+3.77	confirmed	1	1	0	0	1.85	0.83	0	+0	-0.03
A	777	5	482	0.12	4.63	+4.51	confirmed	1	1	0	0	0.70	1.61	0	+0	-0.03
A	777	6	24	0.39	3.96	+3.57	confirmed	1	1	0	0	4.57	0.93	0	-28	+0.08
A	777	7	16	0.48	4.73	+4.25	confirmed	1	0	0	0	9.69	1.17	0	+4	-0.09
A	777	8	289	0.33	6.11	+5.77	pretender	1	1	0	0	2.81	1.04	0	-20	-0.02
A	777	9	327	0.47	6.63	+6.16	pretender	1	1	0	0	6.93	1.26	1	-24	+0.01
A	777	10	101	0.64	11.17	+10.52	pretender	1	1	0	0	3.73	1.11	0	+186	+0.11
A	777	11	249	0.68	7.44	+6.76	pretender	1	1	0	0	6.71	0.71	0	+44	-0.05
A	777	12	25	0.25	6.63	+6.37	pretender	1	1	0	0	6.24	1.18	0	+40	+0.06
A	777	13	37	0.75	2.58	+1.83	confirmed	1	1	0	0	4.63	1.41	0	-4	+0.02
A	777	14	481	0.14	1.97	+1.83	confirmed	1	0	0	0	6.49	0.40	0	-12	-0.04
A	777	15	246	0.69	9.73	+9.04	pretender	1	1	0	0	4.97	1.07	0	-283	-0.02
A	777	16	43	0.16	4.39	+4.23	confirmed	1	1	0	0	3.93	1.14	0	+44	+0.02
A	777	17	333	0.51	7.36	+6.85	pretender	1	1	0	0	2.69	1.33	0	+48	+0.07
A	777	18	312	1.48	5.32	+3.84	pretender	1	1	0	0	7.92	1.18	0	-81	-0.10
A	777	19	6	0.71	5.21	+4.50	pretender	1	1	0	0	9.07	1.21	0	+0	-0.00
A	777	20	413	0.07	3.36	+3.29	confirmed	1	1	1	0	0.76	1.04	0	+0	-0.03
B	42	1	358	0.45	1.84	+1.38	confirmed	1	0	1	0	3.65	0.63	0	+28	-0.03
B	42	2	43	1.14	2.60	+1.46	confirmed	0	1	0	0	3.06	0.37	0	+14	+0.02
B	42	3	24	0.76	3.80	+3.05	confirmed	1	0	1	0	3.14	0.59	0	+28	-0.02
B	42	4	428	0.43	3.76	+3.33	confirmed	1	1	0	0	2.87	0.78	0	+71	+0.04
B	42	5	487	1.07	1.96	+0.88	confirmed	0	0	0	0	1.66	0.21	0	+0	+0.00
B	42	6	396	1.10	1.62	+0.52	confirmed	0	1	0	0	2.93	0.26	0	+14	-0.03
B	42	7	302	0.61	31.64	+31.03	pretender	1	1	0	0	6.10	1.20	1	+933	+0.29
B	42	8	468	0.63	5.86	+5.23	pretender	1	1	0	0	3.90	0.66	0	-339	-0.03
B	42	9	310	0.70	2.78	+2.08	confirmed	1	1	0	0	3.09	0.74	0	-01	-0.01
B	42	10	38	1.88	6.82	+4.95	pretender	1	1	0	0	7.98	0.50	0	+14	+0.10
B	42	11	16	1.03	2.51	+1.48	confirmed	0	1	0	0	2.71	0.39	0	+28	-0.01
B	42	12	168	0.35	7.58	+7.23	pretender	1	1	1	0	2.17	0.87	0	-57	-0.07
B	42	13	339	0.30	2.56	+2.25	confirmed	1	0	0	0	3.39	0.60	0	+42	-0.04

S	seed	#	idx	Surr MAE %	RCWA MAE %	Δ pp	status	T1	T2	T3	T4	Surr@truth %	$\ u^*-u_{true}\ $	support viol.	peak shift nm	peak ΔA
B	42	14	146	0.30	7.54	+7.24	pretender	1	0	0	0	1.89	0.46	0	+28	-0.09
B	42	15	118	0.31	1.49	+1.18	confirmed	1	0	1	0	4.22	0.73	0	-28	+0.04
B	42	16	481	1.22	7.25	+6.03	pretender	1	1	0	0	4.92	0.64	0	+71	+0.02
B	42	17	172	0.37	10.74	+10.37	pretender	1	1	0	0	4.17	0.48	0	+438	+0.21
B	42	18	311	0.67	13.40	+12.72	pretender	1	1	1	0	2.77	0.34	0	-820	-0.06
B	42	19	51	0.49	3.53	+3.04	confirmed	1	1	0	0	1.81	0.71	0	+0	-0.08
B	42	20	255	1.07	13.68	+12.61	pretender	1	1	0	0	1.59	1.68	0	+0	+0.18
B	123	1	358	0.65	7.31	+6.66	pretender	1	0	1	0	4.04	0.85	0	-71	+0.00
B	123	2	43	0.97	4.53	+3.56	confirmed	1	1	0	0	3.53	0.45	0	+28	+0.02
B	123	3	24	0.47	3.77	+3.31	confirmed	1	0	1	0	2.45	0.45	0	+42	+0.01
B	123	4	428	0.56	3.50	+2.94	confirmed	1	1	0	0	3.27	0.51	0	-14	+0.04
B	123	5	487	1.15	3.89	+2.74	confirmed	1	1	0	0	1.36	0.35	0	-99	-0.00
B	123	6	396	0.90	2.91	+2.01	confirmed	1	0	1	0	2.58	0.52	0	+99	-0.03
B	123	7	302	0.56	16.43	+15.87	pretender	1	1	0	0	5.42	1.21	1	+749	+0.11
B	123	8	468	0.39	10.68	+10.29	pretender	1	1	0	0	6.64	1.14	0	+424	+0.21
B	123	9	310	0.61	3.07	+2.46	confirmed	1	1	0	0	2.19	0.88	0	-14	-0.02
B	123	10	38	1.36	5.35	+3.99	pretender	1	1	0	0	10.67	0.83	0	-57	+0.05
B	123	11	16	0.90	6.87	+5.97	pretender	1	1	0	0	2.85	0.43	0	+0	-0.04
B	123	12	168	0.59	25.79	+25.20	pretender	1	1	0	0	2.59	1.24	0	-156	-0.19
B	123	13	339	0.32	3.75	+3.43	confirmed	1	0	0	0	1.99	0.64	0	+0	-0.06
B	123	14	146	0.30	9.42	+9.12	pretender	1	1	0	0	1.05	0.90	0	+57	-0.12
B	123	15	118	0.34	26.96	+26.62	pretender	1	1	1	0	3.04	1.06	0	-636	-0.30
B	123	16	481	1.12	4.50	+3.38	confirmed	1	1	0	0	5.59	0.45	0	+57	+0.08
B	123	17	172	0.42	10.17	+9.74	pretender	1	1	0	0	2.45	0.41	0	+537	+0.13
B	123	18	311	1.02	5.51	+4.50	pretender	1	1	0	0	5.03	1.26	0	-552	+0.13
B	123	19	51	0.53	2.01	+1.47	confirmed	1	1	0	0	2.68	0.72	0	-14	-0.01
B	123	20	255	1.47	17.50	+16.03	pretender	1	1	0	0	2.70	1.37	0	-57	+0.14
B	777	1	358	0.64	1.69	+1.05	confirmed	0	0	1	0	3.99	0.75	0	-14	-0.02
B	777	2	43	1.00	2.08	+1.07	confirmed	0	1	1	0	4.39	0.41	0	+14	-0.01
B	777	3	24	0.45	2.56	+2.11	confirmed	1	0	1	0	3.07	0.73	0	+0	+0.01
B	777	4	428	0.42	4.07	+3.66	confirmed	1	1	0	0	2.90	0.64	0	+0	+0.03
B	777	5	487	1.26	4.44	+3.18	confirmed	1	0	1	0	2.08	0.27	0	-113	-0.00
B	777	6	396	1.00	3.16	+2.16	confirmed	1	0	1	0	2.57	0.21	0	+0	+0.00
B	777	7	302	1.16	12.76	+11.60	pretender	1	1	0	0	4.89	1.31	1	+0	+0.10
B	777	8	468	0.79	12.28	+11.48	pretender	1	0	1	0	6.12	1.03	0	-665	-0.10
B	777	9	310	0.71	2.16	+1.45	confirmed	1	1	0	0	2.22	0.79	0	-14	-0.02
B	777	10	38	1.61	7.03	+5.42	pretender	1	1	0	0	9.30	0.78	1	+28	-0.02
B	777	11	16	0.86	6.27	+5.42	pretender	1	1	1	0	3.27	0.45	0	+0	-0.07
B	777	12	168	0.24	5.52	+5.28	pretender	1	0	0	0	2.63	0.79	0	+1089	+0.00
B	777	13	339	0.30	4.49	+4.19	confirmed	1	1	0	0	1.68	0.53	0	-14	-0.06
B	777	14	146	0.25	5.16	+4.91	pretender	1	1	0	0	1.63	0.50	0	+99	-0.04
B	777	15	118	0.51	33.58	+33.07	pretender	1	1	0	0	4.00	1.22	0	-42	-0.36
B	777	16	481	1.13	4.96	+3.82	confirmed	1	1	0	0	5.83	0.59	0	+71	+0.09
B	777	17	172	0.34	4.64	+4.29	confirmed	1	1	0	0	3.21	0.91	0	-42	+0.06
B	777	18	311	1.11	9.27	+8.16	pretender	1	1	0	0	5.04	1.43	0	-834	-0.06
B	777	19	51	0.80	21.24	+20.44	pretender	1	1	0	0	2.43	0.80	1	+14	-0.32
B	777	20	255	1.02	13.14	+12.13	pretender	1	1	0	0	1.82	1.41	0	+1343	+0.22
C	42	1	363	2.57	9.15	+6.58	pretender	1	1	0	0	5.92	0.84	1	+14/-255	+0.54/+0.05
C	42	2	221	0.50	3.60	+3.10	confirmed	1	0	1	0	2.59	0.76	0	+127/+42	+0.07/+0.02
C	42	3	399	2.16	3.79	+1.63	confirmed	0	0	1	0	6.11	0.63	0	-99/+0	+0.08/+0.01
C	42	4	382	1.75	3.42	+1.67	confirmed	0	0	0	0	2.61	0.18	0	+14/+42	-0.02/+0.00
C	42	5	121	1.09	3.37	+2.28	confirmed	1	0	1	0	3.69	0.25	0	+0/+156	+0.04/+0.06
C	42	6	178	1.75	3.80	+2.05	confirmed	0	0	0	0	2.63	0.46	0	+0/+0	+0.04/-0.00
C	42	7	203	3.88	5.59	+1.71	pretender	0	0	1	0	9.29	0.42	0	+382/+0	+0.08/-0.01
C	42	8	341	0.42	5.72	+5.30	pretender	1	0	1	0	1.10	0.11	0	+113/-354	+0.04/+0.11
C	42	9	139	1.36	6.87	+5.51	pretender	1	1	0	0	2.41	0.28	1	-113/+0	+0.15/-0.02
C	42	10	244	3.29	6.49	+3.20	pretender	0	1	0	0	5.70	0.69	0	+127/+28	+0.00/-0.10
C	42	11	176	3.88	11.06	+7.18	pretender	0	1	0	0	2.56	0.94	0	+42/+156	+0.07/-0.19
C	42	12	271	0.83	5.67	+4.84	pretender	1	0	1	0	3.32	0.50	0	+1174/+0	+0.11/+0.02
C	42	13	245	0.40	10.79	+10.39	pretender	1	1	0	0	1.29	0.28	1	+962/-410	+0.34/-0.04
C	42	14	6	2.79	3.82	+1.03	confirmed	0	1	0	0	3.93	0.18	0	+226/-14	-0.01/+0.01
C	42	15	208	0.87	14.54	+13.67	pretender	1	1	0	0	2.14	0.33	0	+848/-283	+0.29/-0.12
C	42	16	129	2.01	3.99	+1.98	confirmed	0	1	0	0	8.38	1.24	0	-14/-127	-0.02/-0.08
C	42	17	463	3.62	5.70	+2.08	pretender	0	1	0	0	2.32	1.45	1	+28/+156	+0.00/-0.04
C	42	18	137	2.76	9.78	+7.02	pretender	1	1	0	0	6.03	0.96	0	-71/-28	-0.07/-0.01
C	42	19	45	2.13	7.92	+5.78	pretender	1	1	0	0	4.97	0.69	0	+14/-42	+0.03/+0.01
C	42	20	403	0.43	11.93	+11.50	pretender	1	0	1	0	0.54	0.48	0	+1400/+0	+0.37/+0.00
C	123	1	363	2.40	9.22	+6.82	pretender	1	1	1	0	4.50	0.73	1	-14/-71	+0.51/+0.09
C	123	2	221	0.47	3.50	+3.02	confirmed	1	0	1	0	1.35	0.24	0	-14/+85	-0.02/+0.07
C	123	3	399	1.83	2.36	+0.53	confirmed	0	0	1	0	4.87	0.43	0	-57/+0	+0.04/+0.05
C	123	4	382	1.70	3.66	+1.96	confirmed	0	0	0	0	2.77	0.21	0	-14/+57	-0.02/+0.00
C	123	5	121	1.04	6.06	+5.01	pretender	1	1	0	0	3.42	0.48	0	+0/-608	-0.03/-0.00
C	123	6	178	1.80	7.56	+5.76	pretender	1	0	0	0	2.10	0.43	0	+0/+14	+0.07/-0.08
C	123	7	203	3.92	5.52	+1.61	pretender	0	0	1	0	9.41	0.37	0	+382/+0	+0.14/-0.01
C	123	8	341	0.42	6.85	+6.43	pretender	1	0	1	0	0.94	0.19	0	-156/-99	-0.08/-0.09
C	123	9	139	1.89	2.73	+0.84	confirmed	0	1	0	0	2.41	0.79	0	+42/+85	-0.07/+0.04
C	123	10	244	4.06	8.89	+4.84	pretender	0	1	0	0	6.51	1.10	0	-834/-170	-0.01/+0.05
C	123	11	176	3.13	8.02	+4.89	pretender	0	1	1	0	2.44	1.22	0	+368/-28	-0.01/-0.24
C	123	12	271	0.73	4.54	+3.81	confirmed	1	0	0	0	2.73	0.64	0	+0/+1259	+0.01/+0.03
C	123	13	245	0.37	5.56	+5.19	pretender	1	0	1	0	2.20	1.09	1	+424/-438	+0.02/-0.06
C	123	14	6	3.32	3.67	+0.36	confirmed	0	1	1	0	4.44	0.19	0	+226/-28	-0.00/+0.00
C	123	15	208	0.43	2.62	+2.18	confirmed	1	1	1	0	2.31	0.39	0	+42/-113	+0.02/-0.03
C	123	16	129	2.72	7.82	+5.10	pretender	0	1	0	0	7.89	1.25	0	-71/+14	-0.02/+0.16
C	123	17	463	4.20	14.46	+10.26	pretender	1	1	0	0	3.62	1.52	1	-339/+14	+0.40/+0.26
C	123	18	137	3.60	6.80	+3.20	pretender	0	1	0	0	5.62	0.86	0	-28/-57	-0.05/-0.01

S	seed	#	idx	Surr MAE %	RCWA MAE %	Δ pp	status	T1	T2	T3	T4	Surr@truth %	$\ u^*-u_{\text{true}}\ $	support viol.	peak shift nm	peak ΔA
C	123	19	45	2.49	11.26	+8.77	pretender	1	1	0	0	5.38	0.74	0	+71/-42	+0.01/-0.01
C	123	20	403	0.42	2.38	+1.96	confirmed	1	1	1	0	0.72	0.12	0	+0/+0	+0.02/+0.01
C	777	1	363	2.27	7.54	+5.26	pretender	1	0	0	0	3.93	0.75	1	-14/+0	+0.46/+0.09
C	777	2	221	0.47	7.82	+7.35	pretender	1	1	1	0	2.01	1.10	0	+156/+170	+0.10/+0.12
C	777	3	399	2.16	5.93	+3.77	pretender	0	1	1	0	5.84	1.10	0	-113/+0	+0.00/-0.05
C	777	4	382	1.78	3.45	+1.68	confirmed	0	0	0	0	3.09	0.19	0	+42/+28	+0.01/+0.00
C	777	5	121	1.20	4.52	+3.32	confirmed	1	1	1	0	3.89	0.47	0	+0/-622	-0.03/-0.01
C	777	6	178	2.25	5.20	+2.95	pretender	0	0	0	0	2.31	0.41	0	+14/+0	+0.02/+0.03
C	777	7	203	4.66	5.50	+0.84	pretender	0	0	1	0	9.45	0.37	0	+382/+0	+0.10/-0.02
C	777	8	341	0.25	3.71	+3.46	confirmed	1	0	1	0	0.81	0.05	0	-156/+14	-0.09/+0.01
C	777	9	139	1.53	5.08	+3.55	pretender	1	1	0	0	2.86	0.62	0	+14/+0	-0.11/-0.05
C	777	10	244	4.07	5.81	+1.73	pretender	0	0	0	0	6.92	0.72	0	-566/+14	-0.07/-0.06
C	777	11	176	2.69	5.12	+2.42	pretender	0	1	0	0	1.95	0.91	0	+42/+42	+0.04/-0.28
C	777	12	271	0.84	3.70	+2.86	confirmed	1	0	0	0	4.80	0.47	0	+1061/+0	+0.07/+0.01
C	777	13	245	0.48	8.85	+8.37	pretender	1	1	0	0	1.73	0.76	1	-71/-438	+0.35/+0.05
C	777	14	6	2.35	3.84	+1.49	confirmed	0	1	0	0	3.96	0.15	0	+212/-28	+0.01/+0.01
C	777	15	208	1.57	5.68	+4.11	pretender	1	0	1	0	2.44	1.36	0	+71/-297	+0.17/+0.03
C	777	16	129	2.91	—	—	degenerate	0	1	0	0	8.84	1.16	0	—	—
C	777	17	463	3.47	5.08	+1.61	pretender	0	1	1	0	2.10	1.42	0	+42/+127	-0.00/-0.03
C	777	18	137	3.13	10.87	+7.75	pretender	1	1	0	0	5.61	1.36	1	-71/-212	+0.11/-0.04
C	777	19	45	1.90	11.37	+9.46	pretender	1	1	0	0	4.63	0.76	0	+57/-42	+0.02/-0.01
C	777	20	403	0.43	2.60	+2.16	confirmed	1	1	0	0	0.58	0.06	0	+0/+0	-0.01/+0.01

Valid designs: 179; pretenders: 95.

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